

Heat conductance in infinite space

Solve $\partial_t u = D \partial_x^2 u$ (1)

for $x \in \mathbb{R}$, and $u(x, 0) = f(x)$

$$u(x, t) = \int \frac{dk}{\sqrt{2\pi}} e^{i x \cdot k} \hat{u}(k, t)$$

$$\partial_t u(x, t) = \int \frac{dk}{\sqrt{2\pi}} e^{i x \cdot k} \partial_t \hat{u}(k, t)$$

$$\partial_x^2 u(x, t) = \int \frac{dk}{\sqrt{2\pi}} (i k)^2 e^{i x \cdot k} \hat{u}(k, t)$$

$\Rightarrow$ (1) reads

$$\int \frac{dk}{\sqrt{2\pi}} e^{i x \cdot k} (\partial_t \hat{u}(k, t) + D k^2 \hat{u}(k, t)) = 0$$

i.e., the inv. F.T. of $\partial_t \hat{u}(k, t) + D k^2 \hat{u}(k, t)$ is identically 0 $\Rightarrow$

$$\partial_t \hat{u}(k, t) + D k^2 \hat{u}(k, t) = 0$$

This we can solve $\Rightarrow$

$$\hat{u}(k, t) = \hat{u}_0(k) \cdot e^{-D \cdot k^2 \cdot t}$$

solution:

$$u(x, t) = \int \frac{dk}{\sqrt{2\pi}} e^{i \cdot k \cdot x} \hat{u}_0(k) e^{-D \cdot k^2 \cdot t}$$

$\hat{u}_0(k)$ is determined
by initial condition

$$f(x) = u(x, 0) = \int \frac{dk}{\sqrt{2\pi}} e^{i \cdot k \cdot x} \hat{u}_0(k)$$

$$\Rightarrow \hat{u}_0(k) = \hat{f}(k) = \int \frac{dx'}{\sqrt{2\pi}} e^{-i \cdot k \cdot x'} f(x')$$

$$u(x, t) = \int \frac{dk}{\sqrt{2\pi}} e^{i \cdot k \cdot x} \hat{f}(k) e^{-D \cdot k^2 \cdot t}$$

or, alternatively:

$$\begin{aligned} u(x, t) &= \int \frac{dk}{\sqrt{2\pi}} e^{i \cdot k \cdot x} \int \frac{dx'}{\sqrt{2\pi}} e^{-i \cdot k \cdot x'} f(x') e^{-D \cdot k^2 \cdot t} \\ &= \int dx' \underbrace{\left(\int_{-\infty}^{\infty} \frac{dk}{\sqrt{2\pi}} \frac{1}{\sqrt{2\pi}} e^{i \cdot k \cdot (x - x')} e^{-D \cdot k^2 \cdot t} \right)}_{G_t(x - x')} f(x') \end{aligned}$$

G is referred to as Green's function

$$G_t = \mathcal{F}^{-1} \left[\frac{1}{\sqrt{2\pi}} e^{-D \cdot k^2 \cdot t} \right] = \frac{1}{\sqrt{2\pi}} \underbrace{\mathcal{F}^{-1} \left[e^{-\frac{1}{2} k^2 (2 \cdot D \cdot t)} \right]}_{\frac{1}{\sqrt{2Dt}} e^{-\frac{1}{2} \frac{x^2}{2 \cdot D \cdot t}}}$$

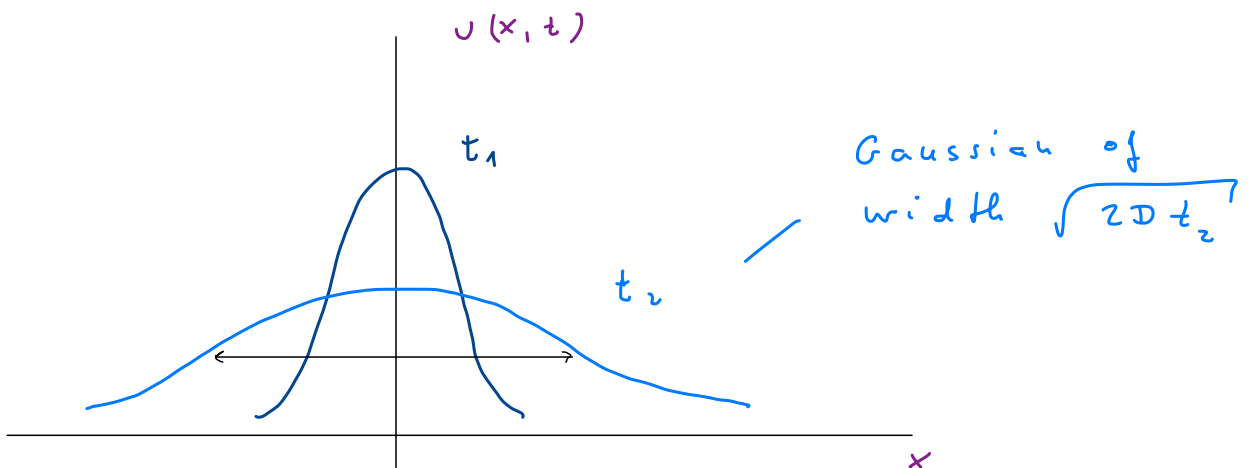
$$G_t(x) = \frac{1}{\sqrt{4\pi \cdot D \cdot t}} e^{-\frac{x^2}{4 \cdot D \cdot t}}$$

$$u(x, t) = \int dx' G_t(x - x') f(x')$$

consider initial condition, $f(x) = \delta(x)$

then:

$$u(x, t) = G_t(x)$$



$$\text{height} \sim \frac{1}{\sqrt{t}}$$